SDBench

We audited Microsoft SDBench on synthetic case 0. Both doctor models we ran, GPT-5 and Astra max, passed with 4/5: each named the lymphoma (IVLBCL) but missed the associated HLH. We found eight issues in how the benchmark gets to that score: what the Gatekeeper reveals or makes up, how it handles grouped orders, and what the Judge checks. Open a finding to see how each setup we tested behaved. None shows incorrect work getting credit in a real run, so there is no P0 so far.

Audited source · 304ce9625a04 ↗ Runs 2026-10-25 (GPT-5) and 2026-10-29 (Astra max) · 1 of 5 synthetic cases PDF

● Seen: the issue showed up ● Not seen: tested, issue absent ○ Not tested in this setup

Skin biopsy returned an immunohistochemistry panel nobody ordered `skin biopsy`

Beyond the request | Requested items not reported P1

● GPT-5 ● Astra max ● Astra Gatekeeper

The skin biopsy is the test that established the diagnosis in this case. What comes back with it should depend on what the doctor ordered.

● GPT-5 run

Doctor GPT-5 · Gatekeeper GPT-5 · observed run, 2026-10-25

GPT-5 ordered deep punch biopsies with H&E, direct immunofluorescence and bacterial/fungal/AFB stains. It did not name CD20, PAX5, Ki-67 or EBER. The Gatekeeper returned the source report's immunohistochemistry findings anyway, while answering the immunofluorescence and stains it did name as "not reported." The response was logged as `is_synthetic: false`.

▾ Request and response

What the doctor requested GPT-5 · /test_logs/53/test_name

```
Dermatology punch biopsies of the mottled/telangiectatic skin lesions (deep punch to
subcutis) for H&E and direct immunofluorescence; include bacterial/fungal/AFB stains
```

What the Gatekeeper returned GPT-5 · /test_logs/53/result · is_synthetic: false

Dermatology deep punch biopsies (livedoid/telangiectatic lesions and clinically normal skin): Epidermis unremarkable. Numerous small vessels in deep dermis and subcutis (capillaries, postcapillary venules, small arterioles) are distended/occluded by cohesive aggregates of large atypical mononuclear cells with scant cytoplasm, vesicular nuclei, prominent nucleoli; mitoses frequent; occasional intraluminal fibrin thrombi. Minimal perivascular inflammation; atypical cells confined intraluminally. Similar findings present in biopsy of normal-appearing skin.

Immunohistochemistry: Atypical intraluminal cells positive for CD20, PAX5, CD79a; negative for CD3 and cytokeratin. Endothelial markers (CD31/CD34) highlight endothelium encasing intraluminal aggregates. Cells express MUM1/IRF4 and BCL2; CD10 negative; BCL6 weak; subset CD5 positive. Ki-67 proliferation index approximately 95%. EBER in-situ hybridization: negative. Direct immunofluorescence and special stains for bacteria/fungi/AFB: not reported.

Raw JSON log


```
{
  "test_name": "Test to order: Dermatology punch biopsies of the mottled/telangiectatic skin lesions (deep punch to subcutis) for H&E and direct immunofluorescence; include bacterial/fungal/AFB stains",
  "reason": "",
  "result": "Dermatology deep punch biopsies (livedoid/telangiectatic lesions and clinically normal skin): Epidermis unremarkable. Numerous small vessels in deep dermis and subcutis (capillaries, postcapillary venules, small arterioles) are distended/occluded by cohesive aggregates of large atypical mononuclear cells with scant cytoplasm, vesicular nuclei, prominent nucleoli; mitoses frequent; occasional intraluminal fibrin thrombi. Minimal perivascular inflammation; atypical cells confined intraluminally. Similar findings present in biopsy of normal-appearing skin.\n\nImmunohistochemistry: Atypical intraluminal cells positive for CD20, PAX5, CD79a; negative for CD3 and cytokeratin. Endothelial markers (CD31/CD34) highlight endothelium encasing intraluminal aggregates. Cells express MUM1/IRF4 and BCL2; CD10 negative; BCL6 weak; subset CD5 positive. Ki-67 proliferation index approximately 95%. EBER in-situ hybridization: negative. Direct immunofluorescence and special stains for bacteria/fungi/AFB: not reported.",
  "is_synthetic": false,
  "timestamp": "2026-10-25T18:41:25.721187",
  "batch_mode": true
}
```


Astra max run

Doctor Astra max · Gatekeeper GPT-5 · observed run, 2026-10-29

Astra named the markers itself: CD20, PAX5, CD3, CD31/CD34 and Ki-67. The same GPT-5 Gatekeeper returned exactly those, under “Immunohistochemistry (requested

markers)", and none of the rest of the panel. So the same Gatekeeper drew the boundary differently in the two runs.

Request and response

What the doctor requested Astra max · /test_logs/14/test_name

```
Obtain three deep skin biopsies, sampling the mottled/telangiectatic areas and clinically normal skin, with each specimen including subcutaneous fat. Request histopathology and immunohistochemistry including CD20, PAX5, CD3, CD31/CD34, and Ki-67, specifically assessing for atypical lymphoid cells within small-vessel lumina.
```

What the Gatekeeper returned GPT-5 · /test_logs/14/result · is_synthetic: false

```
Random deep skin biopsies (thigh livedoid area, trunk telangiectatic patch, and clinically normal upper arm), each extending into subcutaneous fat:

Histopathology: Epidermis unremarkable. Numerous dilated small vessels in the deep dermis and subcutaneous fat (capillaries, postcapillary venules, small arterioles) are distended and largely occluded by cohesive aggregates of large atypical mononuclear cells. The atypical cells are intraluminal; minimal infiltration of surrounding stroma. Mitotic figures are frequent; apoptotic debris present. Fibrin thrombi noted in several involved vessels. Similar intravascular aggregates are present in the specimen from clinically normal-appearing skin.

Immunohistochemistry (requested markers): The intraluminal large cells are strongly positive for CD20 and PAX5; negative for CD3. Endothelial markers CD31 and CD34 highlight a thin rim of endothelium surrounding the intraluminal aggregates, confirming localization within vascular lumina. Ki-67 proliferation index is approximately 95%.
```

Raw: astra-max-diagnosis/json/case_k0_0.json

Astra Gatekeeper

GPT-5's saved order resent to an Astra Gatekeeper · replay, no diagnosis or Judge

Given GPT-5's exact order, an Astra Gatekeeper returned the documented histology without the immunohistochemistry panel. It generated the requested immunofluorescence and stains under a separate "Synthetic ancillary results" label. Each setup ran once, so we can't tell how much of the difference is the model and how much is run-to-run variation.

Astra Gatekeeper response

```
Documented histologic findings: Numerous dilated small vessels in the deep dermis and within the subcutaneous fat — capillaries, postcapillary venules, and small arterioles — were
```

distended and largely occluded by cohesive aggregates of large, atypical mononuclear cells. These cells had scant cytoplasm, large vesicular nuclei with coarse chromatin, and one or more prominent nucleoli; mitotic figures were frequent, and apoptotic debris was present. Admixed fibrin thrombi were noted within several involved vessels. There was minimal infiltration of the surrounding dermal or subcutaneous stroma.
Synthetic ancillary results: Direct immunofluorescence showed no specific vascular or dermoepidermal immune deposits. Bacterial, fungal, and acid-fast stains showed no organisms.

Raw: astra-gatekeeper/replay/comparison.json

The panel is in the source report

Nothing was invented: the full panel, with B-cell markers, Ki-67 of about 95% and a negative EBER, is in the source. The question is scope, meaning which results belong to which order.

Source case report

> Immunohistochemical staining established the lineage and localization of these cells. The atypical intraluminal cells were strongly positive for the B-cell markers CD20, PAX5, and CD79a, and were negative for the T-cell marker CD3 and for cytokeratin, excluding a carcinoma. Endothelial markers, CD31 and CD34, decorated a thin rim of endothelium surrounding the neoplastic aggregates, confirming unequivocally that the malignant cells resided *within* the vascular lumina rather than in the perivascular tissue. The cells expressed MUM1/IRF4 and BCL2 and were negative for CD10, with only weak BCL6 expression, indicating a non–germinal-center (activated B-cell–like) immunophenotype. CD5 was expressed by a subset of cells. The proliferation index, assessed by Ki-67, was very high, at approximately 95%. In-situ hybridization for Epstein–Barr virus–encoded RNA (EBER) was negative.

The Gatekeeper’s instructions pull in two directions

The prompt forbids diagnosis-identifying information beyond what is requested, and it also says to return an existing report verbatim. It does not define which ancillary tests belong to a biopsy order, so this is not a definite rule violation.

Gatekeeper prompt · v6_fixed

> **Prohibited disclosures** – never reveal or imply: ... any information *beyond what is requested* that identifies the diagnosis (e.g. "consistent with X" or "suggestive of X")

> If the investigation already exists in the vignette → return its report verbatim.

Why this may affect evaluation

If only named tests should be returned, the GPT-5 run exceeded scope. If a biopsy order includes the pathology department's follow-up panel, that scope and its cost must apply to every participant. As it stands, the same Gatekeeper gave one doctor the full panel unasked and gave the other only what it asked for. This is not a claim that the tests are clinically inappropriate.

CSF cytology also returned flow cytometry `lumbar puncture`

Beyond the request | Requested items not reported | P1

GPT-5 | Astra max | Astra Gatekeeper

Cytology and flow cytometry are separate CSF tests. Only cytology was ordered.

GPT-5 run

Doctor GPT-5 · Gatekeeper GPT-5 · observed run, 2026-10-25

GPT-5 ordered opening pressure, cell count, protein and glucose, Gram stain and culture, viral PCRs, cryptococcal antigen, fungal and AFB studies and, last, CSF cytology. The Gatekeeper appended `CSF flow cytometry: no clonal lymphoid population.` Meanwhile the requested enterovirus PCR, fungal culture and AFB studies came back "not reported in the current record"; none of them are in the source. The response was logged as `is_synthetic: false`.

Request and response

What the doctor requested GPT-5 · /test_logs/52/test_name

```
Lumbar puncture after neuroimaging: opening pressure; CSF cell count with differential, protein, glucose; Gram stain and bacterial culture; HSV-1/2 PCR; VZV PCR; enterovirus PCR; cryptococcal antigen; fungal culture; AFB smear/culture; CSF cytology
```

What the Gatekeeper returned GPT-5 · /test_logs/52/result · is_synthetic: false

```
Lumbar puncture: Opening pressure 17 cm H2O. CSF: 14 nucleated cells/µL (90% lymphocytes), 2 red cells/µL; protein 76 mg/dL; glucose 64 mg/dL with simultaneous serum glucose 102 mg/dL. Gram stain: no organisms; bacterial cultures: no growth. HSV-1/2 PCR: negative; VZV PCR: negative; cryptococcal antigen: negative. CSF cytology: reactive lymphocytes, no malignant
```

cells. CSF flow cytometry: no clonal lymphoid population. Enterovirus PCR, fungal culture, and AFB smear/culture: not reported in the current record.

Raw JSON log


```
{
  "test_name": "Test to order: Lumbar puncture after neuroimaging: opening pressure; CSF cell count with differential, protein, glucose; Gram stain and bacterial culture; HSV-1/2 PCR; VZV PCR; enterovirus PCR; cryptococcal antigen; fungal culture; AFB smear/culture; CSF cytology",
  "reason": "",
  "result": "Lumbar puncture: Opening pressure 17 cm H2O. CSF: 14 nucleated cells/µL (90% lymphocytes), 2 red cells/µL; protein 76 mg/dL; glucose 64 mg/dL with simultaneous serum glucose 102 mg/dL. Gram stain: no organisms; bacterial cultures: no growth. HSV-1/2 PCR: negative; VZV PCR: negative; cryptococcal antigen: negative. CSF cytology: reactive lymphocytes, no malignant cells. CSF flow cytometry: no clonal lymphoid population. Enterovirus PCR, fungal culture, and AFB smear/culture: not reported in the current record.",
  "is_synthetic": false,
  "timestamp": "2026-10-25T18:41:25.721185",
  "batch_mode": true
}
```


Astra max run

Doctor Astra max · Gatekeeper GPT-5 · observed run, 2026-10-29

Astra never ordered a lumbar puncture. The CSF results, flow cytometry included, reached it through one broad question instead (see the broad-question finding below).

Astra Gatekeeper

GPT-5's saved order resent to an Astra Gatekeeper · replay, no diagnosis or Judge

Given the same order, an Astra Gatekeeper returned the documented CSF results without flow cytometry. For enterovirus PCR, fungal culture and AFB studies it generated negative results, labeled "Synthetic additional results."

Astra Gatekeeper response


```
Documented results: Opening pressure, 17 cm of water. CSF contained 14 nucleated cells per cubic millimeter (90% lymphocytes) and 2 red cells per cubic millimeter; protein, 76 mg/dL; glucose, 64 mg/dL, with simultaneous serum glucose of 102 mg/dL. Gram staining and bacterial cultures were negative, as were PCR assays for herpes simplex virus and varicella-zoster virus and a cryptococcal antigen assay. CSF cytologic examination revealed reactive lymphocytes but no malignant cells.
Synthetic additional results: Enterovirus PCR negative. Fungal culture showed no growth after
```

```
final incubation. AFB smear negative; mycobacterial culture showed no growth after final
incubation.
```

Raw: astra-gatekeeper/replay/comparison.json

The flow cytometry result is in the source report

The source reports CSF cytology and flow cytometry in the same sentence. The Gatekeeper returned both for an order that named one.

Source case report

> A diagnostic lumbar puncture was performed. The opening pressure was 17 cm of water. The cerebrospinal fluid (CSF) contained 14 nucleated cells per cubic millimeter (90% lymphocytes) and 2 red cells per cubic millimeter; the protein level was 76 mg/dL (reference range, 15–45) and the glucose level was 64 mg/dL with a simultaneous serum glucose of 102 mg/dL. Gram staining and bacterial cultures were negative, as were polymerase-chain-reaction assays for herpes simplex virus and varicella–zoster virus and a cryptococcal antigen assay. CSF cytologic examination revealed reactive lymphocytes but no malignant cells, and flow cytometry of the CSF identified no clonal lymphoid population.

Why this may affect evaluation

The doctor receives a result for a test it did not select. If the benchmark measures test-level information gathering and cost, what is disclosed should match the actual order. If CSF testing is treated as a package, its components and price should be fixed in advance.

Urine culture returned a negative result that isn't in the source `urine culture`

Not in source | Source and generated mixed | P1

● GPT-5 ○ Astra max ● Astra Gatekeeper

The source report has a urinalysis but no urine culture. For anything missing, the prompt tells the Gatekeeper to generate a plausible result and flag it.

● **GPT-5 run**

Doctor GPT-5 · Gatekeeper GPT-5 · observed run, 2026-10-25

GPT-5 ordered a urinalysis and urine culture. The Gatekeeper returned the urinalysis from the source plus a generated negative culture. The whole response carries a single `is_synthetic: true` flag, so it doesn't say which line was generated.

Request and response

What the doctor requested GPT-5 · /test_logs/29/test_name

```
Urinalysis and urine culture
```

What the Gatekeeper returned GPT-5 · /test_logs/29/result · is_synthetic: true

```
Urinalysis: Specific gravity 1.015; trace protein; no glucose; no leukocyte esterase or
nitrites; 0–2 RBC/HPF; 0–2 WBC/HPF; no casts.
Urine culture: No growth at 48 hours.
```

Raw JSON log


```
{
  "test_name": "Test to order: Urinalysis and urine culture",
  "reason": "",
  "result": "Urinalysis: Specific gravity 1.015; trace protein; no glucose; no leukocyte
esterase or nitrites; 0–2 RBC/HPF; 0–2 WBC/HPF; no casts.\nUrine culture: No growth at 48
hours.",
  "is_synthetic": true,
  "timestamp": "2026-10-25T18:41:25.721160",
  "batch_mode": true
}
```


○ Astra max run

Doctor Astra max · Gatekeeper GPT-5 · observed run, 2026-10-29

Astra did not order a urine culture. The urinalysis reached it through the broad question, without a culture.

● Astra Gatekeeper

GPT-5's saved order resent to an Astra Gatekeeper, then split · replay, no diagnosis or Judge

An Astra Gatekeeper refused the combined order. Sent as two requests, it returned the urinalysis from the source and generated a negative culture, flagged synthetic.

Astra Gatekeeper responses

```
Request: Urinalysis
Response: The urinalysis showed a specific gravity of 1.015, trace protein, no glucose, no
leukocyte esterase or nitrites, and 0–2 red cells and 0–2 white cells per high-power field,
without casts.
is_synthetic: false

Request: Urine culture
Response: Urine culture: No growth after 48 hours.
is_synthetic: true
```

Raw: astra-gatekeeper/replay/split-requests/comparison.json

The source has blood cultures, not a urine culture

The source's "three sets of blood cultures" are blood cultures. No urine culture is reported anywhere.

Source case report

> The urinalysis showed a specific gravity of 1.015, trace protein, no glucose, no leukocyte esterase or nitrites, and 0–2 red cells and 0–2 white cells per high-power field, without casts. Three sets of blood cultures obtained over 48 hours showed no growth. Testing for human immunodeficiency virus (HIV) antigen and antibody was negative; hepatitis B surface antigen, hepatitis B core antibody, and hepatitis C antibody were negative; and a rapid plasma reagin test was nonreactive. Antinuclear antibody, anti–double-stranded DNA antibody, antineutrophil cytoplasmic antibodies (both proteinase-3 and myeloperoxidase specificities), rheumatoid factor, and anti–cyclic citrullinated peptide antibody were all negative or within normal limits. The serum C3 and C4 concentrations were normal. Cryoglobulins were not detected. A lupus anticoagulant assay was negative, and anticardiolipin and anti–β2-glycoprotein I antibodies (IgG and IgM) were within normal limits. Serum and urine protein electrophoresis with immunofixation showed no monoclonal protein. Serologic testing for *Bartonella henselae*, *Coxiella burnetii*, *Brucella* species, and *Borrelia burgdorferi* was negative, as was a tuberculosis interferon-γ release assay.

The current configuration allows this

Generation is what the prompt asks for, so it is not evidence of an implementation error. The design question is how the validity and consistency of generated results are established.

Gatekeeper prompt · v6_fixed

> Else → fabricate a plausible synthetic result and set `<is_synthetic>true</is_synthetic>`.

Why this may affect evaluation

A generated negative culture could lead a doctor model to deprioritize a urinary source of fever. These runs do not establish that it changed any decision.

Anaplasma and Ehrlichia tests returned generated negatives `Anaplasma · Ehrlichia`

Not in source

P1

● GPT-5 ○ Astra max ● Astra Gatekeeper

The source lists other infection serologies but nothing on Anaplasma or Ehrlichia, two tick-borne infections.

● GPT-5 run

Doctor GPT-5 · Gatekeeper GPT-5 · observed run, 2026-10-25

GPT-5 ordered PCR and serology for both organisms. The Gatekeeper returned “PCR: Not detected” and negative IgG and IgM for each, flagged `is_synthetic: true`.

Request and response

What the doctor requested GPT-5 · /test_logs/43/test_name

```
Anaplasma and Ehrlichia PCR/serology
```

What the Gatekeeper returned GPT-5 · /test_logs/43/result · is_synthetic: true

```
Anaplasma phagocytophilum:
- PCR: Not detected
- IgG: Negative
- IgM: Negative
Ehrlichia chaffeensis:
- PCR: Not detected
- IgG: Negative
- IgM: Negative
```

Raw JSON log

```
{
  "test_name": "Test to order: Anaplasma and Ehrlichia PCR/serology",
  "reason": "",
  "result": "Anaplasma phagocytophilum:\n- PCR: Not detected\n- IgG: Negative\n- IgM:
Negative\nEhrlichia chaffeensis:\n- PCR: Not detected\n- IgG: Negative\n- IgM: Negative",
  "is_synthetic": true,
  "timestamp": "2026-10-25T18:41:25.721175",
  "batch_mode": true
}
```


○ Astra max run

Doctor Astra max · Gatekeeper GPT-5 · observed run, 2026-10-29

Astra did not order these tests. It asked about tick exposure instead, and the answer, “no recent tick bites reported”, matches the source.

● Astra Gatekeeper

GPT-5’s saved order resent to an Astra Gatekeeper, then split · replay, no diagnosis or Judge

An Astra Gatekeeper refused the combined order. Asked as six separate tests, it generated negative PCR, IgG and IgM results for both organisms, each flagged synthetic. Its answers add titers and reference ranges that GPT-5’s did not; nothing in the benchmark fixes what a generated result should say.

▾ Astra Gatekeeper responses


```
Request: Anaplasma PCR
Response: Anaplasma phagocytophilum PCR, whole blood: Not detected.
is_synthetic: true

Request: Anaplasma IgG serology
Response: Anaplasma phagocytophilum IgG by immunofluorescence: <1:64 (negative; reference:
<1:64).
is_synthetic: true

Request: Anaplasma IgM serology
Response: Anaplasma phagocytophilum IgM antibody: <1:20 (negative; reference: <1:20).
is_synthetic: true

Request: Ehrlichia PCR
Response: Ehrlichia species PCR, whole blood: Not detected.
is_synthetic: true

Request: Ehrlichia IgG serology
```

```
Response: Ehrlichia chaffeensis IgG antibody: <1:64 (negative; reference: <1:64).
is_synthetic: true

Request: Ehrlichia IgM serology
Response: Ehrlichia chaffeensis IgM antibody: <1:20 (negative; reference: <1:20).
is_synthetic: true
```

Raw: astra-gatekeeper/replay/split-requests/comparison.json

What the source does list

The source lists serologies for Bartonella, Coxiella, Brucella and Borrelia and a tuberculosis interferon-γ release assay. Anaplasma and Ehrlichia appear nowhere in it. The prompt allows generating the missing results.

Source case report

> … Serum and urine protein electrophoresis with immunofixation showed no monoclonal protein. Serologic testing for *Bartonella henselae*, *Coxiella burnetii*, *Brucella* species, and *Borrelia burgdorferi* was negative, as was a tuberculosis interferon-γ release assay.

Why this may affect evaluation

Generated negatives could lead a doctor model to drop tick-borne infections from its differential, and two Gatekeepers phrase them differently. These runs do not establish that they changed any decision.

One broad question released the admission workup, including HLH markers `patient question`

Beyond the request | Suspected rule conflict | P1

○ GPT-5 ● Astra max ○ Astra Gatekeeper

Patient questions are for history and symptoms, but one question can also ask for results the doctor would otherwise have to order test by test.

○ **GPT-5 run**

Doctor GPT-5 · Gatekeeper GPT-5 · observed run, 2026-10-25

GPT-5's 18 questions covered symptoms, history and exposures; none asked for prior results. In its run, the triglycerides, sIL-2R and marrow findings never reached the doctor.

Astra max run

Doctor Astra max · Gatekeeper GPT-5 · observed run, 2026-10-29

In its first turn Astra asked nine questions. The ninth asked which investigations had already been performed and what they showed. The GPT-5 Gatekeeper answered in one reply with the labs, CSF and initial bone-marrow results, all from this admission, including ferritin 3,150 ng/mL, triglycerides 295 mg/dL, soluble IL-2 receptor 4,250 U/mL and marrow hemophagocytosis. It held back only the current-admission imaging, which the prompt says must stay hidden until ordered; the prompt has no matching rule for labs or marrow results. The response was logged as `is_synthetic: false`.

Astra still left HLH out of its final answer, which named IVLBCL with CNS, skin and bone-marrow involvement. The Judge took a point off for the missing HLH and the missing non–germinal-center qualifier (4/5). So the HLH miss was not caused by missing information.

Question and response

What the doctor asked Astra max · first turn, question 9

```
What investigations have already been performed, including blood counts, liver and kidney
tests, cultures, brain imaging, or lumbar puncture, and what were the results?
```

What the Gatekeeper returned GPT-5 · /test_logs/8/result · is_synthetic: false

```
Investigations to date:
- Hematology: Hgb 9.4 g/dL, Hct 28.5%, MCV 89 fL, WBC 3.6×10⁹/L, platelets 96×10⁹/L,
reticulocytes 1.0%; smear: normochromic normocytic anemia, no schistocytes, occasional
teardrop forms, no blasts.
- Coagulation: PT 13.1 s, aPTT 31 s, fibrinogen 360 mg/dL, D-dimer 2.9 µg/mL FEU.
- Chemistry/inflammation: AST 76 U/L, ALT 52 U/L, alkaline phosphatase 168 U/L, total
bilirubin 1.1 mg/dL, albumin 3.0 g/dL, LDH 842 U/L, haptoglobin 15 mg/dL, ferritin 3,150
ng/mL, triglycerides 295 mg/dL, uric acid 7.8 mg/dL, ESR 88 mm/hr, CRP 96 mg/L, procalcitonin
0.12 ng/mL, soluble IL-2 receptor 4,250 U/mL, β2-microglobulin 4.2 mg/L, lactate 1.4 mmol/L,
thyrotropin 2.3 µIU/mL.
- Urinalysis: specific gravity 1.015; trace protein; 0–2 RBC and 0–2 WBC per HPF; no
leukocyte esterase or nitrites; no casts.
- Microbiology/serology: three sets of blood cultures—no growth. HIV antigen/antibody
negative; hepatitis B surface antigen and core antibody negative; hepatitis C antibody
negative; RPR nonreactive. Serologies negative for Bartonella henselae, Coxiella burnetii,
Brucella spp., and Borrelia burgdorferi. TB interferon-γ release assay negative.
- Autoimmune panel: ANA, anti-dsDNA, ANCA (PR3/MPO), rheumatoid factor, anti-CCP
negative/within normal limits; C3/C4 normal; cryoglobulins not detected; lupus anticoagulant
negative; anticardiolipin and anti-β2-glycoprotein I (IgG/IgM) within normal limits;
serum/urine protein electrophoresis with immunofixation—no monoclonal protein.
```

```
- Lumbar puncture: opening pressure 17 cm H₂O; CSF 14 nucleated cells/mm³ (90% lymphocytes),
2 RBC/mm³; protein 76 mg/dL; glucose 64 mg/dL (serum 102 mg/dL). CSF Gram stain/cultures
negative; HSV and VZV PCR negative; cryptococcal antigen negative; cytology with reactive
lymphocytes; flow cytometry—no clonal lymphoid population.
- ADAMTS13 activity: 68% (reference >60%).
- Imaging performed during this admission (reports available upon order): CT
chest/abdomen/pelvis with contrast; MRI brain with and without contrast; intracranial and
cervical MRA; transthoracic and transesophageal echocardiography; whole-body FDG PET-CT.
- Bone marrow: aspirate/biopsy hypercellular for age with trilineage hematopoiesis; increased
activated macrophages with hemophagocytosis; no overt sheets of malignant cells on initial
aspirate; flow cytometry—no clearly abnormal population.
```

Raw: astra-max-diagnosis/json/case_k0_0.json · API: api_calls/gatekeeper_001.json

○ **Astra Gatekeeper**

GPT-5's saved orders resent to an Astra Gatekeeper · replay, no diagnosis or Judge

The replay resent test orders only, so no patient question was tested.

Diagnostic findings arrived without a single order

The prompt says findings that may determine the diagnosis should only come from targeted tests, and that patient questions which would break this rule must be refused.

Gatekeeper prompt · v6_fixed

> Only allow the disclosure of findings that may determine the diagnosis in targeted and specific tests (such as microbiology or viral cultures growing specific organisms, biopsies showing specific cell types, genetic tests revealing specific genetic mutation). Otherwise refuse to disclose the finding and ask the agent to order the specific test.

> If answering would break the Pathognomonic safeguard → refuse with the standard template.

The prompt also allows chart lookups

Another rule lets the Gatekeeper answer from the chart when the answer is there. For earlier non-imaging investigations, the prompt allows details only if they are non-diagnostic. It does not say how broad a "what has been done" question may be, or which findings count as diagnosis-determining, so this is a suspected violation, not a confirmed one.

Gatekeeper prompt · v6_fixed

> If the answer exists in the chart → quote it in clinical-documentation style (no first-person quotes).

> **Historic investigations (non-imaging)** – May be acknowledged by type + date; provide details only if documented and non-diagnostic.

Why this may affect evaluation

Astra placed 7 test orders against GPT-5's 36, but much of its information came from this one question, and several of its orders cover more than one test or specimen. Comparing order counts or test costs between the two runs would be misleading.

Bartonella IgG and IgM were each reported negative, though the source gives one result `Bartonella serology`

P1

Labeled as source | Suspected over-specification

GPT-5 · Astra max · Astra Gatekeeper

`is_synthetic` is the benchmark's only record of where a result came from. Details inferred beyond the source shouldn't be labeled as source.

GPT-5 run

Doctor GPT-5 · Gatekeeper GPT-5 · observed run, 2026-10-25

GPT-5 ordered henselae and quintana IgG/IgM with optional PCR in one order. The Gatekeeper reported each as negative but flagged the whole response synthetic, which covers the inferred parts.

Request and response

```
Request: Bartonella henselae and quintana serology (IgG/IgM) ± PCR
Response: Bartonella henselae IgG: Negative; IgM: Negative
Bartonella quintana IgG: Negative; IgM: Negative
Bartonella PCR (blood): Not detected
is_synthetic: true
```

Raw: gpt5-trajectory/case_k0_0.json · /test_logs/40

Astra max run

Doctor Astra max · Gatekeeper GPT-5 · observed run, 2026-10-29

Astra did not order Bartonella tests. The broad-question answer said serologies were negative for Bartonella henselae, matching the source's wording.

Astra Gatekeeper

Split requests sent to an Astra Gatekeeper · replay, no diagnosis or Judge

In the split test (see the order-handling finding), two requests asked for Bartonella henselae IgG and IgM serology. Both answers were "Negative" with `is_synthetic: false`.

Requests and responses

```
Request: Bartonella henselae IgG serology
Response: Bartonella henselae IgG serology: Negative.
is_synthetic: false

Request: Bartonella henselae IgM serology
Response: Bartonella henselae IgM serology: Negative.
is_synthetic: false
```

Raw: astra-gatekeeper/replay/split-requests/comparison.json

An inference presented as a documented result

Turning "serology negative" into "IgG negative" and "IgM negative" is a reasonable inference, but the source doesn't state it. With the flag set to false, a reader would take both as documented. We have not shown that either value is clinically wrong, or which panel was actually run.

Source case report

> … Serum and urine protein electrophoresis with immunofixation showed no monoclonal protein. Serologic testing for *Bartonella henselae*, *Coxiella burnetii*, *Brucella* species, and *Borrelia burgdorferi* was negative, as was a tuberculosis interferon-γ release assay.

Why this may affect evaluation

If inferred details are labeled as source, generated and documented information can't be told apart. A safer answer would be "serology negative; individual IgG/IgM results not stated."

An Astra Gatekeeper refused 10 grouped orders but answered the same tests one at a time `grouped orders`

P1

Answer depends on order wording | Cause not isolated

GPT-5 · Astra max · Astra Gatekeeper

The prompt requires each order to name a single orderable investigation, but gives no list of what counts as one.

GPT-5 run

Doctor GPT-5 · Gatekeeper GPT-5 · observed run, 2026-10-25

The GPT-5 Gatekeeper answered all 36 orders, including the 10 grouped ones, such as "Rheumatoid factor (RF) and anti-CCP".

Example: RF and anti-CCP

```
Request: Rheumatoid factor (RF) and anti-CCP
Response: Rheumatoid factor: Within normal limits.
Anti-cyclic citrullinated peptide (anti-CCP): Within normal limits.
is_synthetic: false
```

Astra max run

Doctor Astra max · Gatekeeper GPT-5 · observed run, 2026-10-29

The GPT-5 Gatekeeper answered all seven of Astra's orders, several of which cover more than one test or specimen, such as both echocardiograms, or three skin biopsies with immunostains.

Astra Gatekeeper

GPT-5's 36 saved orders resent to an Astra Gatekeeper, then the refused ones split · replay, no diagnosis or Judge

An Astra Gatekeeper (`openai/gpt-6-astra`, provider-default reasoning) answered 26 of the 36 orders and refused 10 with the prompt's standard refusal, which gives no reason. We split those 10 into 31 requests, one order each and mostly single tests. It answered all 31

and flagged 17 as synthetic. It refused “Rheumatoid factor (RF) and anti-CCP” as one order but answered each separately from the source. Yet it accepted the skin biopsy with several stains as one order, and in the split test “ENA panel”, which we had deliberately kept whole, was answered as one request with a generated six-antibody result.

▾ The 10 refused orders

```
 1. Urinalysis and urine culture
 2. Thyroid-stimulating hormone (TSH) and free T4
 3. Serum protein electrophoresis (SPEP) with immunofixation and serum free light chains
 4. Antinuclear antibody (ANA) with ENA panel and anti-double-stranded DNA
 5. Rheumatoid factor (RF) and anti-CCP
 6. Cryoglobulins and cryofibrinogen
 7. Hepatitis B surface antigen, core antibody; Hepatitis C antibody
 8. Bartonella henselae and quintana serology (IgG/IgM) ± PCR
 9. Anaplasma and Ehrlichia PCR/serology
10. Babesia blood smear and PCR
```

Refusal text Astra · same for all 10 · is_synthetic: false

```
I can only provide specific test results or answer specific patient questions. Please request
a specific diagnostic test or ask a specific question about the patient's history, symptoms,
or demographics.
```

▾ RF and anti-CCP, asked separately

```
Request: Rheumatoid factor (RF)
Response: Rheumatoid factor: negative or within normal limits.
is_synthetic: false

Request: Anti-CCP antibody
Response: Anti-cyclic citrullinated peptide antibody: within normal limits.
is_synthetic: false
```

Raw: replay/comparison.json · split-requests/comparison.json

No list defines a single test

The prompt requires a single orderable investigation and refuses umbrella orders, but doesn’t list what counts as one test. Splitting also changed how many requests went into

each call, so we can't separate the two effects. Each condition ran once.

Gatekeeper prompt · v6_fixed

> Must name a single orderable investigation performed by a department *unless* the order concerns **routine bedside observations** (see below).

> Refuse umbrella orders (e.g. "all infection tests", "complete blood work").

Why this may affect evaluation

With an Astra Gatekeeper, whether a doctor model gets a result can depend on how it words and groups its orders, not only on which tests it chooses. This does not prove inconsistent enforcement: a biopsy with stains or a named panel may reasonably count as one test.

A false explanation in the trace still scored 5/5 `Judge probe`

Process not graded | Constructed answer — P1

GPT-5 Judge · Astra max Judge

We built this test ourselves; no doctor model wrote the false statement. Starting from the saved Astra run, we replaced the final answer with the exact reference diagnosis (A). Then we added a false statement to the skin test log's reason field, which the Judge reads (C). Each Judge graded A and C three times. No doctor or Gatekeeper calls were made.

The planted statement

```
The skin biopsy showed no atypical intravascular cells.
```

The actual biopsy result reports atypical intravascular cells. Putting the statement in the doctor's prose before the diagnosis tag (B) changes nothing, because the Judge never sees that text. Raw: judge-probes/false-intermediate-step/comparison.json

GPT-5 Judge

3 grades per answer

A: 5, 5, 5. C: 5, 5, 5. None of its grading notes explicitly mentioned the contradiction.

Astra max Judge

3 grades per answer

A: 5, 5, 5. C: 5, 5, 5. In one of its C grades it explicitly noted the contradiction and still gave 5/5.

Its grading note

> Although the skin-biopsy order contains a contradictory negative-result parenthetical, the **actual reported result** and pathological narrative document diagnostic intravascular malignant B cells.

In a second probe, the Judge did catch an unsupported addition

We added an unsupported “secondary hypogammaglobulinemia” to the correct answer. Both Judges dropped it to 3/5 in all three repeats, while the correct answer got 5/5 each time.

The two answers

```
A · correct answer
Intravascular large B-cell lymphoma (non-germinal-center B-cell immunophenotype), with
central-nervous-system and cutaneous involvement and associated secondary hemophagocytic
lymphohistiocytosis.

B · with an unsupported addition
Intravascular large B-cell lymphoma (non-germinal-center B-cell immunophenotype), with
central-nervous-system and cutaneous involvement and associated secondary hemophagocytic
lymphohistiocytosis. The lymphoma is also complicated by secondary hypogammaglobulinemia.
```

Raw: judge-probes/unsupported-complication/comparison.json

Why this may affect evaluation

The score certifies the final diagnosis, not the reasoning or steps behind it. That matches the rubric’s stated scope, so this is a scope limit rather than a grading bug. It does mean a passing score says nothing about whether the trace is truthful.

Benchmark and grading details

How SDBench is scored

For each case, SDBench runs three roles in a loop. The doctor model receives a short case summary that does not reveal the diagnosis and, turn by turn, orders tests, asks questions or commits to a diagnosis. The Gatekeeper answers each order from the case's full text. When the case contains the result, it returns that; when it does not, its prompt tells it to generate a synthetic result, though in the GPT-5 run it sometimes answered "not reported" instead. When the doctor commits, a Judge scores the diagnosis against the ground truth on a 1–5 scale; 4–5 counts as correct.

The Gatekeeper and Judge are fixed by the configuration; only the doctor model is being evaluated. So whatever the Gatekeeper discloses feeds straight into the doctor's score.

What we ran

Case	`synthetic` dataset · train · case 0 — "Case 17-2026: A 58-Year-Old Woman with Recurrent Fever, Progressive Confusion, and Multifocal Cerebral Infarction". Reference diagnosis: intravascular large B-cell lymphoma (non–germinal-center B-cell immunophenotype), with central-nervous-system and cutaneous involvement and associated secondary hemophagocytic lymphohistiocytosis.
GPT-5 run · 2026-10-25	Doctor GPT-5 (SDBench baseline agent), Gatekeeper GPT-5, Judge GPT-5; prompts `v6_fixed` / `v4_reasoning_with_tests`. 18 patient questions, then 36 test orders in one turn, then the diagnosis: IVLBCL with skin and CNS involvement. 4/5; the Judge noted the missing non–germinal-center qualifier and HLH. 10 of 36 test responses flagged synthetic.
Astra max run · 2026-10-29	Doctor `openai/gpt-6-astra` with max reasoning (same baseline agent); Gatekeeper and Judge GPT-5 with the same prompts. 9 patient questions, then 7 test orders, then the diagnosis: IVLBCL with CNS, skin and bone-marrow involvement. 4/5 for the same two omissions; the Judge accepted the marrow involvement as correct. None of the 16 responses flagged synthetic.
Astra Gatekeeper replay	The GPT-5 run's 36 saved orders resent unchanged to `openai/gpt-6-astra` as Gatekeeper (provider-default reasoning), then the 10 refused ones split into 31 requests. No doctor turns, no Judge.
Judge probes	Answers we wrote, each graded three times by GPT-5 and by Astra max. No doctor or Gatekeeper calls.

Reading is_synthetic

`is_synthetic` is a label the Gatekeeper model sets itself; it is not an independent check. `false` means the response was not marked as containing fabricated results. It does not certify accuracy, appropriate scope or real-patient provenance, and refusals carry `false` too. `true` means generated content is included, not necessarily that the whole response was generated. The source report is itself synthetic.

What this audit supports

This is not evidence that either doctor model assumed normal results without ordering tests. The models asked and ordered; the Gatekeeper supplied extra information or generated missing results. These logs alone cannot establish how much that affected the final diagnoses.

Both models named the core disease

Each named IVLBCL after the skin-pathology results arrived, and both lost a point for missing HLH. For GPT-5, the B-cell markers that establish the lineage came only through the unrequested panel; Astra ordered that panel itself.

No P0 so far

No real run showed incorrect work receiving credit. The false-explanation result comes from an answer we wrote.

The information-gathering process is not validated

The configured GPT-5 Gatekeeper added unrequested results, answered a broad question with diagnostic findings and generated missing results. With an Astra Gatekeeper, whether an order was answered depended on how it was grouped; the GPT-5 Gatekeeper refused none of the same 36 orders.

The Gatekeeper may influence evaluation

Establishing a causal effect on a final answer would need a separate experiment, such as rerunning with the extra information removed.

A token-saving motive cannot be established

The agent prompt sets no token objective, and the log records no per-test rationale: the harness never fills the `reason` field. We do not infer the models' intent.

Conditions for a consistent comparison

1. Define the returned findings and included follow-up panels for each test ID.
2. Define what a patient question may return, and apply the targeted-test rule to it.
3. Align the test components disclosed to the doctor with those included in costing.
4. Return "unavailable" for missing results, or use reviewed, fixed values shared across participants.

5. Record source-backed versus generated provenance for each test component, not only for the combined response.

Cost estimate

We also priced the GPT-5 run's tests and the physician workup described in the synthetic report with the same automated method: GPT-5 Mini matching tests to a hospital charge master, with LLM estimates where matching failed. Estimated test costs were $21,488 for the physician path and $13,245 for GPT-5 ($23,288 and $13,845 including visit fees). The matching is unverified; for example, GPT-5's MRA/CTA may be counted twice and the physician's skin-biopsy procedure code is missing. We draw no conclusion from these numbers.

Sources and reproducibility

- GPT-5 run trajectory: gpt5-trajectory/case_k0_0.json (SHA-256 `19cdc9815b028e97db4fa9609f12fde44c670761079e84806f302f0ecceb1222`)
- Astra max run: trajectory, Gatekeeper API call, run config
- Astra Gatekeeper: replay of 36 orders, 31 split requests
- Judge probes: false explanation, unsupported complication (each grading call under `runs/`)
- Case report and Gatekeeper prompt: case 0, v6_fixed prompt

This is an information-flow review of saved runs on one synthetic case. It is not an expert clinical assessment, a controlled comparison with human physicians, or a rerun with selected information removed. Each condition ran once unless stated. The original diagnoses and cost results are unchanged.